

LeakBench-Tab: Technical Supplement

Anonymous submission

INTERNAL DRAFT — RESULTS BLOCKED.
This supplement contains no submission-ready empirical result until the final confirmatory claim release generates the evidence macros.

Scope and Artifact Contract

This supplement covers the corrected controlled benchmark and five fixed real-data case studies. The controlled design contains 20 tasks, 11 mechanisms, 5 strengths, 5 model families, and 5 injection seeds (27,500 planned cells). Metadata localization and governance experiments are not part of the confirmatory claim set and are therefore not reported as paper findings. A separate 22,000-cell matrix compares 4 oracle-blind, pre-test development rankers on the same frozen tasks; diagnostic rows are not counted as model-training cells.

Numerical macros are generated only from `results/corrected_v2/paper_claims.json`. The generator verifies a complete confirmatory matrix, final claim statuses, finite ordered intervals, the five-model identity set, all 22,000 diagnostic cells, and evidence provenance before writing LaTeX. Its source digest is PENDING.

Mechanism Contracts and Construction Invariants

Every injected task records a contaminated-field mask, timestamps, entity and source identifiers, feature-availability metadata, structural groups, graph edges, and immutable task and split hashes. C is fixed by these construction contracts before diagnostic or model fitting.

Strength Mapping

The five generic strength labels map to $\{0.2, 0.4, 0.6, 0.8, 1.0\}$. Mechanisms with discrete structural strength use frozen mechanism-specific mappings: M06 redundant-field counts $\{1, 2, 4, 6, 8\}$, M07 subgroup prevalences $\{0.05, 0.10, 0.20, 0.35, 0.50\}$, and M11 component sizes $\{2, 3, 4, 6, 8\}$. M04, M05, M08, and M09 must be documented with their exact future-window, shrinkage, support, and source-assignment parameters from the frozen corrected protocol rather than reconstructed from result tables. M09 represents one semantic source variable with

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eight complete one-hot columns. Its AP evaluates encoded-column localization against those eight oracle-marked columns; M09 D is therefore representation-conditional and is not evidence that a source field is equally detectable under arbitrary encodings or field-level scoring rules.

M04 and M05 use row-count windows after stable timestamp sorting, not fixed calendar durations. For a row at time t , both begin strictly after every row with timestamp t (a right-sided search), so the current row and all same-timestamp rows are excluded. M04 then averages the next $\max(2, \lfloor 0.05n \rfloor)$ rows. M05 first skips $\max(1, \lfloor n\delta \rfloor) - 1$ strictly future rows and averages the next $\max(2, \max(1, \lfloor n\delta \rfloor))$ rows, with the frozen $\delta = 0.02$ at every strength. A future timestamp tie can be truncated by the row-count width. Consequently these mechanisms are sensitive to sampling density and stable order within a future tie; they represent controlled row-window contamination rather than an invariant duration-based event-time estimand. If a row has no eligible strictly future observations after the declared skip, the constructed future statistic falls back to the leave-one-out global label prior; the current row's label is never reintroduced by that fallback.

Controlled Task Registry and Splits

The task registry contains twenty deterministic panel generators across five archetypes. Generator randomness is independent of injection randomness. Rows are ordered chronologically and assigned once to 60% train, 20% validation, and 20% test partitions. The split is invariant to mechanism, strength, and downstream learner. The final camera-ready appendix should list for each task: task identifier, archetype, row and feature counts, entity and source support, class prevalence, generator seed, and split hash. These values must be imported from the canonical manifest, not manually transcribed.

Model Identity and Preprocessing

The core set comprises logistic regression, random forest (Breiman 2001), CatBoost (Prokhorenkova et al. 2018), LightGBM (Ke et al. 2017), and the official TabM implementation (Gorishniy, Kotelnikov, and Babenko 2025). All scaling and learned preprocessing use training rows only. Strict and permissive views share row partitions, labels, optimization seeds, and model hyperparameters. The official

IDs	Category	Required invariant	Construction audit
M01, M02	Simple	Target-derived field; strength changes signal/noise	Separation by strength
M06	Simple	Redundant fields form one declared contaminated group	Group size matches level
M10	Simple	Legitimate component equals clean input exactly	Equality plus separate mask
M03	Boundary	Nonlinear target interaction	Interaction and label audit
M07	Boundary	Contamination restricted to declared subgroup	Prevalence audit
M11	Boundary	Contaminated fields form declared graph component	Component audit
M04, M05	Structured	Current and past rows excluded from future statistic	Temporal index audit
M08	Structured	Future entity outcome; current label excluded	Leave-one-out/prior audit
M09	Structured	Outcome-dependent source collection; complete one-hot	Source-ID permutation invariance

Table 1: Construction invariants that must hold independently of empirical effect size. Final release tables should cite the corresponding automated audit and frozen task-manifest hash.

TabM run uses immutable serialized task bundles and verifies bundle, task, and split hashes before fitting. The release must include package versions, CUDA and GPU identity, determinism settings, optimizer, learning rate, batch size, epoch and early-stopping rules, and per-model preprocessing. Surrogate implementations and silent fallbacks are prohibited.

Estimands and Statistical Procedures

For a downstream learner m , the paired effect is

$$X_m = \text{AUROC}_{m,\text{permissive}} - \text{AUROC}_{m,\text{strict}}. \quad (1)$$

For a feature-localization diagnostic, normalized average precision is

$$D = \frac{\text{AP} - \pi}{1 - \pi}, \quad (2)$$

where π is contaminated-column prevalence. Mutual-information scores are computed from training rows once per injected task with fixed `random_state=42` and must be identical across all five downstream learners. Three additional oracle-blind rankers—absolute correlation, logistic coefficient magnitude, and random-forest permutation importance—form a frozen sensitivity suite. MI, absolute correlation, and logistic coefficients use training rows only. The random forest is fitted on training rows, but permutation importance is evaluated against frozen validation labels. Thus all four are pre-test development diagnostics, not four training-only diagnostics. D is always labeled by diagnostic and encoding; a per-mechanism maximum is not a deployable selector.

The independent unit is the controlled task. The 20,000-draw hierarchical bootstrap first samples tasks with replacement and then samples injection seeds within each selected task. Task-level sign flips test mechanism effects, with Holm adjustment over the eleven-mechanism family. The three declared category contrasts form a separate Holm family. Model-specific simple-minus-structured contrasts form a five-model family. The category tests are exact two-sided sign flips over the 20 task effects, not bootstrap-tail p -values. M08 samples one entity vector per dataset and applies it jointly to all 125 seed-by-model-by-strength cells. It retains a separate effect for each injection seed under that shared entity draw before outer dataset/seed resampling. M09 samples one source-category vector per dataset/seed/strength and applies it jointly to all five models. Its eight source IDs are

designed categories rather than draws from a source population, so the resulting interval is descriptive reweighting only. All failed cells remain in the raw ledger; incomplete matrices are not analyzed as confirmatory.

The diagnostic sensitivity matrix crosses twenty tasks, eleven mechanisms, five strengths, five seeds, and four rankers, for 22,000 cells. The first three rankers use training rows only; RF permutation trains on training rows and evaluates perturbations on the frozen validation split. Intervals use the same task/seed hierarchy as the primary D analysis. Method comparisons are descriptive sensitivity analyses; no paired simultaneous method-comparison interval was frozen, and no ranker is chosen after observing contamination labels.

Frozen diagnostic RNG amendment. After the confirmatory sensitivity suite was complete, an evidence-chain audit found a provenance mismatch: its mutual-information implementation used the injection seed, whereas the frozen headline implementation and the immutable task manifest used fixed seed 42. The post-unblinding amendment is deterministic and scope-locked: it replaces the AP, normalized-AP, and top-5 recall fields on every and only the 5,500 mutual-information rows with the identity-matched values already stored in the frozen task manifest. It preserves every original field on the other 16,500 rows and does not change the four-method set. The amendment builder reads no model-outcome columns and performs no value-dependent row selection or tuning. The task-seeded raw suite is retained only for provenance; the paper and release validator consume the amended canonical diagnostic table. All diagnostic-method and mechanism-profile comparisons are descriptive; the amendment does not authorize thresholded profile conclusions.

Frozen statistical inference amendment. A separate post-unblinding methods audit identified three invalid uncertainty constructions in the superseded outputs: category bootstrap tail areas had been labeled as p -values, D-X intervals resampled X while holding D fixed, and cluster draws were independent across cells generated from the same task. A second audit found that the initial cluster amendment still drew entities independently across M08 injection seeds that reuse the same test rows, labels, and entities. Before final amended analysis, the second replacement code and immutable task registry were hash-frozen.

Learner	Train-fitted preprocessing	Frozen fit configuration	Selection / prediction
LR	StandardScaler	$C = 1$, max 2,000 iterations	Test positive-class probability
RF	None	250 trees, min leaf 2, max features $\sqrt{p}$	Test positive-class probability
CatBoost	None	250 iterations, rate 0.05, depth 6, CPU	Validation AUC; patience 30
LightGBM	None	250 trees, rate 0.05, 31 leaves, depth 6, min child 20	Validation binary log loss; patience 30
TabM	StandardScaler	Official TabM v0.0.3, $k = 32$, AdamW, rate 0.003, weight decay 10^{-4}	Validation BCE; max 200 epochs, patience 20, batch 256; mean of per- k sigmoid probabilities

Table 2: Frozen learner configurations. Every strict/permissive pair shares the split, seed, validation rule, and learner configuration. CPU learners use one thread; official TabM uses deterministic PyTorch algorithms on the declared CUDA device.

The replacement uses exact task-level sign flips with Holm adjustment for the three category contrasts, jointly resamples D and X with the same dataset/seed indices, and synchronizes entity/source-category draws at the grouping levels above. M03, M08, M09, D–X, diagnostic-method, and model-specific summaries remain descriptive; only the pre-existing simple-versus-structured directional contrast uses the exact Holm-and-CI support rule. Output manifests bind analysis code, canonical data, configuration, all twenty frozen task bundles, and every consumed M08/M09 prediction-file hash; row IDs, labels, entities, sources, clean/full AUC, and task hashes are checked directly. The release validator deep-runs every paper-facing statistical table.

After all protocol, amendment, support-rule, and claim-scope freezes, a local hyperparameter search unintentionally displayed a truncated subset of in-progress GPU checkpoint rows. The incident is retained in the artifact audit trail; it authorized no task exclusion, mechanism change, inferential change, threshold change, or manuscript result wording. All reported values are derived only after completion from the fully validated canonical release.

Complete Controlled Results

Complete controlled results are intentionally absent from this draft. No pilot estimate is a substitute for the final canonical matrix.

Diagnostic-Conditional Detectability

Diagnostic sensitivity values remain blocked. In particular, no pilot method-ordering is transcribed into the manuscript.

Cluster Sensitivity for M08 and M09

Cluster-sensitive estimates remain blocked pending complete prediction-file coverage for all confirmatory M08/M09 cells.

Detectability–Exploitability Diagnostics

Association diagnostics remain blocked and are not populated with pilot values.

Real-Data Boundary Audits

The five fixed cases are UCI Bank Marketing (UCI Machine Learning Repository 2014), a local Lending Club accepted-loan mirror, U.S. flight delays (U.S. Bureau of Transportation Statistics 2026), Chicago food inspections (City of Chicago

2026), and NYC 311 (City of New York 2026). The final lineage table must bind the train-fitted-category preprocessing v2 protocol: category vocabularies are learned from training rows, unseen validation/test categories map to a reserved unknown value, and globally encoded date strings are dropped. The earlier full-table-vocabulary outputs are superseded. The lineage table must include source filename and SHA-256, row-selection rule, chronological sort field, prediction boundary, contaminated fields, sample size, split rule, and all adapter exclusions. Lending Club uses accepted applications only; its post-origination servicing, last-credit-pull, hardship, and payment-plan fields are post-boundary. The flight-delay boundary is immediately before scheduled departure: only an explicit schedule allowlist is legitimate, while all actual departure, arrival, and diversion fields are post-event. Bank Marketing uses the canonical full dataset; no synthetic fallback is an evaluable real-data row. These declarations are adapter contracts rather than labels inferred from the observed paired harm.

The Bank Marketing record is CC BY 4.0 and the canonical `bank-additional-full` table contains 41,188 rows. BTS, Chicago, and NYC entries cite their official government portals. The Lending mirror `accepted_2007_to_2018Q4` has no verified upstream provenance or license in the current artifact: it is local-only and must not be redistributed unless the authors supply and verify both.

Real-data numerical summaries remain blocked pending the same final claim release as the controlled results.

Negative, Superseded, and Out-of-Scope Evidence

Legacy eight-model summaries, surrogate neural-model outputs, the pre-bundle TabM partial run, old 10,083/4,032-cell ledgers, and earlier natural-transfer tables are superseded evidence and cannot source this paper. Corrected results must live under the independent corrected-v2 namespace with checksums. A negative result for the pre-existing directional category claim remains reportable as `NOT_SUPPORTED`; a `PENDING`, `REFUTED`, `INTEGRITY_HOLD`, or incomplete claim cannot generate a paper macro.

Frozen M10 protocol amendment. The original M10 strict-view constructor incorrectly removed the injected legitimate duplicate together with the fields marked by `leakage_mask`; the strict and permissive views therefore did not share the same legitimate base. The frozen

amendment defines the strict matrix as `task.X[:, ~leakage_mask]`. The original 2,000 CPU and 500 TabM M10 rows are superseded and replaced under this corrected pairing. Canonical validation excludes every original M10 run identifier and still requires exactly 27,500 model-training cells. Superseded M10 rows cannot source a claim, table, or figure.

Operational metadata and governance modules are engineering/audit work outside the main confirmatory claim set. Their omission is not evidence for or against their efficacy.

Claim–Evidence Matrix

The final release should map each main-paper statement to (i) a stable claim identifier, (ii) its scope and, only for the directional category contrast, its support rule, (iii) final status, (iv) exact metric fields, (v) canonical source hashes, and (vi) allowed and prohibited wording. The paper generator consumes only the six controlled claim IDs and the case-study-only natural block. Every other state is excluded by design.

Complete Machine-Generated Tables

Complete registry, mechanism, model, diagnostic, strength, natural-case, and claim-scope tables are blocked with the numerical results. Pilot tables are not rendered in their place.

Reproducibility Checklist and Ledger

The archival package should provide the frozen corrected protocol, task-bundle manifest, canonical 27,500-cell table, raw failure-preserving CPU and official TabM tables, retained M08/M09 predictions, the raw and amended canonical diagnostic tables, the diagnostic RNG amendment freeze and impact audit, the statistical-amendment freeze and input/output hash manifests, natural-task lineage, all statistics tables, `paper_claims.json`, generated macro source hash, environment locks, and a release validator. Reproduction commands must distinguish CPU and official GPU runs and must never regenerate task bundles in a different dependency environment.

Limitations and Ethical Considerations

The benchmark uses synthetic controlled tasks, designed mechanisms, binary classification, and a limited model set. The real-data component contains five selected boundary audits rather than a random task sample. Leakage findings can reveal risky data practices; releases should avoid exposing credentials, personal information, or restricted raw data. Task manifests and derived statistics should be public only when their source licenses permit it.

References

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